

Transfer operator

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In [mathematics](#), the **transfer operator** encodes information about an [iterated map](#) and is frequently used to study the behavior of [dynamical systems](#), [statistical mechanics](#), [quantum chaos](#) and [fractals](#). In all usual cases, the [largest eigenvalue is 1](#), and the [corresponding eigenvector is the invariant measure](#) of the system.

The transfer operator is sometimes called the **Ruelle operator**, after [David Ruelle](#), or the **Perron–Frobenius operator** or **Ruelle–Perron–Frobenius operator**, in reference to the applicability of the [Perron–Frobenius theorem](#) to the determination of the [eigenvalues](#) of the operator.

The iterated function to be studied is a map $f\colon X\rightarrow X$ for an [arbitrary set](#) X .

The transfer operator is defined as an [operator](#) $\mathcal{L}$ [acting on the space of functions](#) $\{\Phi\colon X\rightarrow\mathbb{C}\}$ as

$$(\mathcal{L}\Phi)(x)=\sum_{y\in f^{-1}(x)}g(y)\Phi(y)$$

where $g\colon X\rightarrow\mathbb{C}$ is an auxiliary valuation function. When f has a [Jacobian](#) determinant $|J|$, then g is usually taken to be $g=1/|J|$.

The above definition of the transfer operator can be shown to be the point-set limit of the measure-theoretic [pushforward](#) of g : in essence, the transfer operator is the [direct image functor](#) in the category of [measurable spaces](#). The left-adjoint of the Perron–Frobenius operator is the [Koopman operator](#) or [composition operator](#). The general setting is provided by the [Borel functional calculus](#).

As a general rule, the transfer operator can usually be interpreted as a (left-) [shift operator](#) acting on a [shift space](#). The most commonly studied shifts are the [subshifts of finite type](#). The adjoint to the transfer operator can likewise usually be interpreted as a right-shift. Particularly well studied right-shifts include the [Jacobi operator](#) and the [Hessenberg matrix](#), both of which generate systems of [orthogonal polynomials](#) via a right-shift.

Whereas the [iteration of a function](#) f naturally leads to a study of the orbits of points of X under iteration (the study of [point dynamics](#)), the transfer operator defines how (smooth) maps evolve under iteration. Thus, transfer operators typically appear in [physics](#) problems, such as [quantum chaos](#) and [statistical mechanics](#), where attention is focused on the time evolution of smooth functions. In turn, this has medical applications to [rational drug design](#), through the field of [molecular dynamics](#).

It is often the case that the transfer operator is positive, has discrete positive real-valued [eigenvalues](#), with the [largest eigenvalue being equal to one](#). For this reason, the transfer operator is sometimes called the Frobenius–Perron operator.

The [eigenfunctions](#) of the transfer operator are usually fractals. When the logarithm of the transfer operator

corresponds to a quantum [Hamiltonian](#), the eigenvalues will typically be very closely spaced, and thus even a very narrow and carefully selected [ensemble](#) of quantum states will encompass a large number of very different fractal eigenstates with non-zero [support](#) over the entire volume. This can be used to explain many results from classical statistical mechanics, including the irreversibility of time and the increase of [entropy](#).

The transfer operator of the [Bernoulli map](#) $b(x) = 2x - \lfloor 2x \rfloor$ is exactly solvable and is a classic example of [deterministic chaos](#); the discrete eigenvalues correspond to the [Bernoulli polynomials](#). This operator also has a continuous spectrum consisting of the [Hurwitz zeta function](#).

The transfer operator of the Gauss map $h(x) = 1/x - \lfloor 1/x \rfloor$ is called the [Gauss–Kuzmin–Wirsing \(GKW\) operator](#). The theory of the GKW dates back to a hypothesis by Gauss on [continued fractions](#) and is closely related to the [Riemann zeta function](#).

[Bernoulli scheme](#)

[Shift of finite type](#)

[Krein–Rutman theorem](#)

[Transfer-matrix method](#)

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